

Nick Monteleone

San Francisco, CA | +1 (719) 466-0135 | nicholasmonteleone25@gmail.com | linkedin.com/in/monteleonenicholas

Summary: *Technically driven Software Engineer with 2 years of product development experience and 7+ years of automation, data systems, and applications engineering in the Biotechnology industry.*

Work Experience

Bhrigu Tech / JSat Automation, Inc

Full-stack software engineer driving backend architecture and implementation, supporting frontend features as needed; transitioned into product development after leading automation engineering efforts on projects for numerous industry clients.

Software Engineer, 2024 - 2025

- Architected and developed FastAPI backend for a recipe manager application, enabling clients to configure control system procedures; integrated with a user-facing chatbot powered by LLMs for natural language interaction.
- Built GraphQL backend in FastAPI for a validation system; leveraged factory-based testing for regression coverage.
- Migrated internal and client applications to new Azure infrastructure after team's transition from JSat to Bhrigu; set up CI/CD pipelines using Azure DevOps, Container Registry, and Virtual Machines.
- Converted existing backend applications from Flask to FastAPI, improving scalability and performance in product suite.
- Validated backend functionality through unit, integration, and end-to-end testing; used AI tools to streamline test generation.
- Developed and deployed internal-use Python packages for shared functionality, including a complex control system file parser.
- Created custom JavaScript-based visualization symbol to display industrial tag data via third-party web API integration.

Senior Automation Engineer, 2022 - 2023

- Automated form editing, documentation, and testing workflows using Python for client workflow application project; containerized applications with Docker to streamline deployment.
- Gathered and documented user, functional, and design requirements for client workflow and training applications; led validation testing to ensure full alignment with specifications prior to release.
- Consulted on and contributed to TypeScript codebase for client displays, helping transition to HTML5-based visualization.
- Led project to virtualize a multi-server control system for a biotechnology firm, delivering validated infrastructure on schedule.

BioMarin Pharmaceutical, Inc.

Dependable and driven engineer supporting automation and data systems for critical manufacturing operations across both commercial and clinical products. Delivered capital project implementations, infrastructure upgrades, and ongoing operations support.

Automation Engineer II, 2020 - 2022

- Installed and tested software applications for a new site-wide distributed control system, including simulation environment.
- Developed documentation, system drawings, and validation protocols for a control system upgrade within downtime window.
- Led bioreactor system reconfiguration by updating controller logic and mapping to accommodate evolving lab requirements.

Data Systems & Applications Engineer I/II, 2018 - 2020

- Created standardized Review-by-Exception batch reports through collaboration with manufacturing and quality teams, aligning recipe alarms with risk assessments; presented solution at an industry-wide automation conference.
- Performed database creations, infrastructure upgrades, and report implementation under GMP change control processes.

Manufacturing Sciences & Technology Intern, 2017

- Designed and implemented a new process monitoring tool by integrating VBA into control system displays and organizing key data points into user-friendly hierarchies, resulting in 23 new visualizations and improved operational efficiency.
-

Education

Rithm School, Full Stack Web Development Program, 2023

University of Colorado, Boulder, 2018, B.S.E Chemical & Biological Engineering, Minor: Biomedical Engineering

Skills

Languages/Frameworks: Python, JavaScript, TypeScript, SQL, FastAPI, Flask, Django, React, Angular, Node.js, Express

Testing: pytest, unittest, Jest, Vitest, selenium

DevOps/Tools: Docker, Git (GitHub & GitLab), Azure DevOps, Azure Container Registry, Azure VMs, AWS S3, WSL, Linux (Ubuntu)

Other: PostgreSQL, SQLite, GraphQL, CI/CD, REST APIs, pandas/numpy, AI tools (Copilot, ChatGPT, Claude, Perplexity)